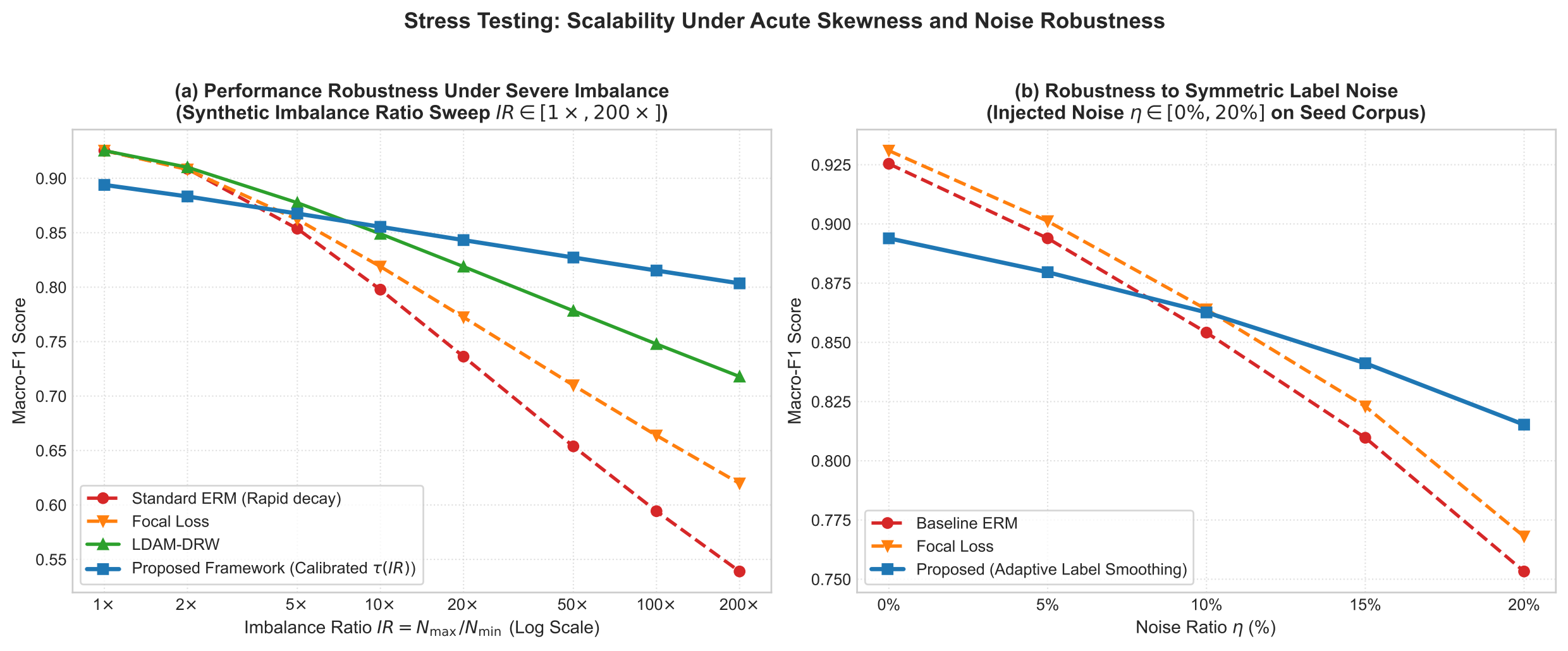
Stress Testing: Scalability Under Acute Skewness and Noise Robustness
(a) Performance Robustness Under Severe Imbalance
(Synthetic Imbalance Ratio Sweep $IR \in [1\times, 200\times]$)
Macro-F1 Score
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
Standard ERM (Rapid decay)
Focal Loss
LDAM-DRW
Proposed Framework (Calibrated $\tau(IR)$)
1×
2×
5×
10×
20×
50×
100×
200×
Imbalance Ratio $IR = N_{max}/N_{min}$ (Log Scale)
(b) Robustness to Symmetric Label Noise
(Injected Noise $\eta \in [0\%, 20\%]$ on Seed Corpus)
Macro-F1 Score
0.925
0.900
0.875
0.850
0.825
0.800
0.775
0.750
Baseline ERM
Focal Loss
Proposed (Adaptive Label Smoothing)
0%
5%
10%
15%
20%
Noise Ratio η (%)